Ans. (b) : Let the number of question is test be X
number of solved question = 4

$\frac{4}{X} \times 100 = 40 \Rightarrow \times = 10$

$\therefore$ The correct answer is option (b).

73. A father is nine times as old as his son and the mother is eight times as old as the son. The sum of the father's and the mother's age is 51 years. What is the age of the son?
(a) 7 years (b) 5 years
(c) 4 years (d) 3 years

Ans. (d) :
Let, the present age of father be F.
The present age of mother be M and son be S.
F = 9S
M = 8S
9S + 8S = 51
17S = 51
S = 3
$\therefore$ The correct answer is option (d).

74. Four persons A, B, C and D consisting of two married couples are in a group. Both the women are shorter than their respective husbands. A is the tallest among the four. C is taller than B. D is B's brother. In this context, which one of the following statements is *not* correct?
(a) All four have family ties.
(b) B is the shortest among the four.
(c) C is taller than D.
(d) A is B's husband.

Ans. (c) :

Husband	Wife
A	B
D	C

- A is the tallest it mean A must be husband.
- D is brother of B it means husband of the other will be D.
- Since, it is given that the height of the wife is less than that of the husband.

Hence, as the statement the height of wife C. Cannot be more than that of husband (D).
$\therefore$ The correct answer is option (C).

GENERAL MATHEMATICS NUMERAL ABILITY

75. Two pipes A and B can independently fill a tank completely in 20 and 30 minutes respectively. If both the pipes are opened simultaneously, how much time will they take to fill the tank completely?
(a) 10 minutes (b) 12 minutes
(c) 15 minutes (d) 25 minutes

Ans. (b) :
Let capacity of tank be 60 litres (LCM of 20 and 30).
Pipe A can fill 3 litres in a minutes. Pipe B can fill 2 litres in one minutes. Both pipes can fill 5 litres in one minute.

Time required $= \frac{60}{5} = 12$ minutes

$\therefore$ The correct answer is option (c).

76. The monthly incomes of Peter and Paul are in the ratio of 4 : 3. Their expenses are in the ratio of 3 : 2. If each saves ₹ 6,000 at the end of the month, their monthly incomes respectively are (in ₹)
(a) 24,000 and 18,000 (b) 28,000 and 21,000
(c) 32,000 and 24,000 (d) 34,000 and 26,000

Ans. (a) :
Income = 4x : 3x
Expense = 3y : 2y
Saving = Income – Expense = 6000
4x – 3y = 6000 and 3x – 2y = 6000
4x – 3y = 3x – 2y
x = y = 6000
$\therefore$ The monthly salary will be 24,000 and 18,000.

77. Two cities A and B are 360 km apart. A car goes from A to B with a speed of 40 km/hr and returns to A with a speed of 60 km/hr. What is the average speed of the car?
(a) 45 km/hr (b) 48 km/hr
(c) 50 km/hr (d) 55 km/hr

Ans. (b) :
Since, the distance covered is the same, the average speed $= \frac{2S_1S_2}{S_1+S_2} = \frac{2\times40\times60}{40+60} = 48$ km/hr.

78. In a 500 metres race, B starts 45 metres ahead of A, but A wins the race while B is still 35 metres behind. What is the ratio of the speeds of A to B assuming that both start at the same time?
(a) 25 : 21 (b) 25 : 20
(c) 5 : 3 (d) 5 : 7

Ans. (a) :
A = 500
B = 500 – 45 – 35 = 420
Since, total time taken by A and B are same.
$\therefore$ The ratio of speed of A and B = 500 : 420 = 25 : 21

79. Two equal glasses of same type are respectively 1/3 and 1/4 full of milk. They are then filled up with water and the contents are mixed in a pot. What is the ratio of milk and water in the pot?
(a) 7 : 17 (b) 1 : 3
(c) 9 : 21 (d) 11 : 23

Ans. (a) :
Let the volume of each glass be 60 litres

Glass A	Glass B
Milk = 20 litres	Milk = 15 litres
Water = 40 litres	Water = 45 litres

Required Ratio $= \frac{\text{Milk}}{\text{Water}} = \frac{35}{85} = \frac{7}{17} = 7:17$

80. An automobile owner reduced his monthly petrol consumption when the prices went up. The price-consumption relationship is as follows :

Price (in ₹ per litre)	40	50	60	75
Monthly consumption (in litres)	60	48	40	32

If the price goes up to ₹ 80 per litre, his expected consumption (in litres) will be
(a) 30 (b) 28
(c) 26 (d) 24

Ans. (a) : The product of price (in Rs./litre) and the monthly consumption (in litres) in constraint is equal to 2400.
Expected consumption when the price goes upto ₹ 80 per litre $= \frac{2400}{80} = 30$ litres.

UNION PUBLIC SERVICE COMMISSION

Civil Services (Preliminary Exam) - 2016

CSAT : PAPER-II

(Chapterwise Analysis with Explanation)

Time : 2 hours **Maximum Number : 200**

Directions for the following 6 (six) items :
Read the following two passages and answer the items that follow each passage. Your answers to these items should be based on the passages only.

Passage-1

Accountability, or the lack of it, in governance generally, and civil services, in particular, is a major factor underlying the deficiencies in governance and public administration. Designing an effective frame work for accountability has been a key element of the reform agenda. A fundamental issue is wither civil services should be accountable to the political executive of the day or to society at large. In other words, how should internal and external accountability be reconciled? Internal accountability is sought to be achieved by internal performance monitoring, official supervision by bodies like the Central Vigilance Commission and Comptroller and Auditor General and judicial review of executive decisions. Articles 311 and 312 of the Indian Constitution provide job security and safeguards to the civil services, especially the All India Services. The framers of the Constitution had envisaged that provision of these safeguards would result in a civil service that is not totally subservient to the political executive but will have the strength to function in larger public interest. The need to balance internal and external accountability is thus built into the Constitution. The issue is where to draw the line. Over the years, the emphasis seems to have tilted in favor of greater internal accountability of the civil services to the political leaders of the day who in turn are expected to be externally accountable to the society at large through the election process. This system for seeking accountability to Society has not worked out, and has led to several adverse consequences for governance.

Some special measures can be considered for improving accountability in civil services. Provisions of articles 311 and 312 should be reviewed and laws and regulations framed to ensure external accountability of civil services. The proposed Civil Services Bill seeks to address some of these requirements. The respective roles of professional civil Services Bill seeks to address some of these requirements. The respective roles of professional civil services and the political executive should he defined so that professional managerial functions and management of civil services are depoliticized. For this purpose, effective statutory civil service boards should be created at the centre and in the states. Decentralization and devolution of authority to bring government and decision making closer to the people also helps to enhance accountability.

1. According to the passage, which of the following factor/factors led to the adverse consequences for governance/public administration?
1. Inability of civil services to strike a balance between internal and external accountabilities
2. Lack of sufficient professional training to the officers to all India Services .
3. Lack of Proper service benefits in civil services
4. Lack of Constitutional provisions to define the respective roles of professional civil services vis-a –vis political executive in this context

Select the correct answer using the code given below :
(a) 1 only
(b) 2 and 3 only
(c) 1 and 4 only
(d) 2, 3 and 4

Ans. (c) : According to the passage, the inability of the civil services to strike a balance between internal and external accountability and the absence of constitutional provisions defining their respective roles in the professional civil services as compared to the political executive.

2. With reference to the passage, the following assumptions have been made:
1. Political executive is an obstacle to the accountability of the civil services to the society
2. In the present framework of Indian polity, the Political executive is no longer accountable to the society

Which of these assumptions is /are valid ?
(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Ans. (a) : Assumption 1 states that political executive hinders the working of civil services and its role in public interest .
This is correct because the emphasis is on internal accountability to the political leaders and the larger public interest is undermined.

3. Which one of the following is the essential massage implied by this passage ?
(a) Civil services are not accountable to the society they are serving
(b) Educated and enlightened persons are not taking up political leadership